Chapter 16, Problem 99P

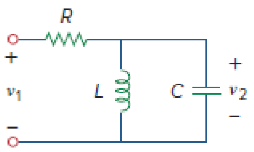
Problem

It is desired to realize the transfer function

$$\frac{V_2(s)}{V_1(s)} = \frac{2s}{s^2 + 2s + 6}$$

using the circuit in Fig. 16.108. Choose R = 1 kΩ and find L and C.

Figure 16.108:



Step-by-step solution

Step 1 of 2

$$SL \,||\, \frac{1}{SC} = \frac{SL.\frac{1}{SC}}{SL + \frac{1}{SC}} = \frac{SL}{1 + S^2LC}$$

$$\frac{V_2}{V_1} = \frac{\frac{SL}{1+S^2LC}}{R + \frac{SL}{1+S^2LC}} + \frac{SL}{S^2RLC + SL + R}$$

$$\frac{V_2}{V_1} = \frac{S.\frac{1}{RC}}{S^2 + \frac{S}{RC} + \frac{1}{LC}}$$

Step 2 of 2

Comparing this with the given transfer function,

$$2 = \frac{1}{RC} \qquad 6 = \frac{1}{LC}$$

$$If \qquad R = 1K\Omega \qquad C = \frac{1}{2R} = 500\,\mu F$$

$$L = \frac{1}{6C} = 333.3\,H$$